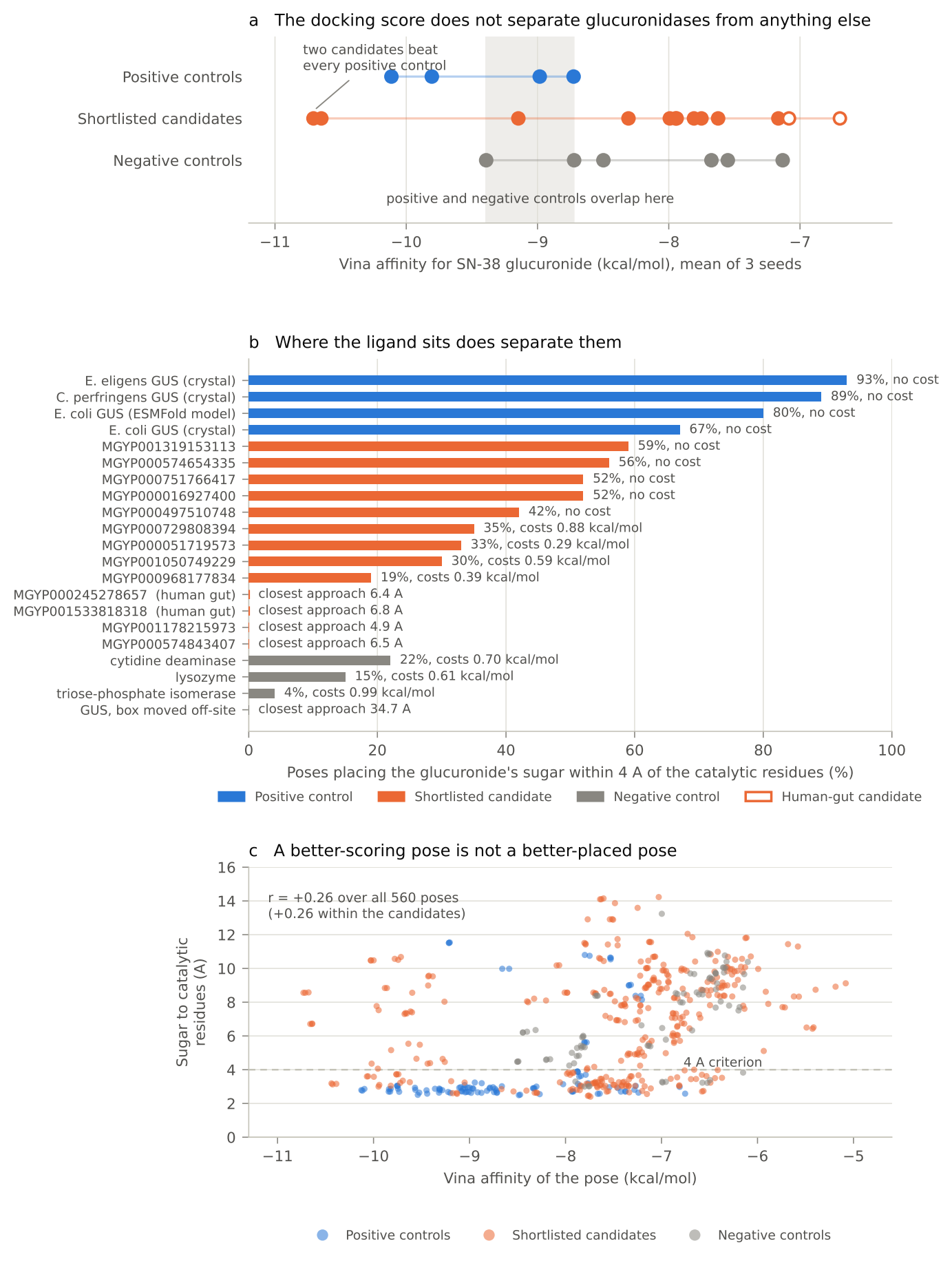
a The docking score does not separate glucuronidases from anything else
two candidates beat every positive control
Positive controls
Shortlisted candidates
Negative controls
positive and negative controls overlap here
−11
−10
−9
−8
−7
Vina affinity for SN-38 glucuronide (kcal/mol), mean of 3 seeds
b Where the ligand sits does separate them
E. eligens GUS (crystal)
C. perfringens GUS (crystal)
E. coli GUS (ESMFold model)
E. coli GUS (crystal)
MGYP001319153113
MGYP000574654335
MGYP000751766417
MGYP000016927400
MGYP000497510748
MGYP000729808394
MGYP000051719573
MGYP001050749229
MGYP000968177834
MGYP000245278657 (human gut)
MGYP001533818318 (human gut)
MGYP001178215973
MGYP000574843407
cytidine deaminase
lysozyme
triose-phosphate isomerase
GUS, box moved off-site
93%, no cost
89%, no cost
80%, no cost
67%, no cost
59%, no cost
56%, no cost
52%, no cost
52%, no cost
42%, no cost
35%, costs 0.88 kcal/mol
33%, costs 0.29 kcal/mol
30%, costs 0.59 kcal/mol
19%, costs 0.39 kcal/mol
closest approach 6.4 A
closest approach 6.8 A
closest approach 4.9 A
closest approach 6.5 A
22%, costs 0.70 kcal/mol
15%, costs 0.61 kcal/mol
4%, costs 0.99 kcal/mol
closest approach 34.7 A
0
20
40
60
80
100
Poses placing the glucuronide's sugar within 4 A of the catalytic residues (%)
Positive control
Shortlisted candidate
Negative control
Human-gut candidate
c A better-scoring pose is not a better-placed pose
r = +0.26 over all 560 poses
(+0.26 within the candidates)
Sugar to catalytic residues (A)
4 A criterion
0
2
4
6
8
10
12
14
16
−11
−10
−9
−8
−7
−6
−5
Vina affinity of the pose (kcal/mol)
Positive controls
Shortlisted candidates
Negative controls